

Bhaskar Pardeshi

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Summary

Seeking Ph.D. internship opportunities in Computer Science with a focus on datacenter networks and operating systems. Specializing in network congestion control, network load balancing, end-host network stacks, compute overload control, and CPU scheduling.

Education

- Georgia Institute of Technology** Atlanta, USA
Ph.D. in Computer Science, CGPA: 4.0/4.0 Aug 2023 – Present
- College of Engineering, Pune** Pune, India
B.Tech in Computer Engineering, CGPA: 9.6/10 Aug 2017 – Jun 2021

Publications

- Svalinn: Overload Control in Large-Scale Servers with Multiple Resource Bottlenecks**
Bhaskar Pardeshi, Peidi Song, Ahmed Saeed *USENIX OSDI 2026*
- CoreSync: A Protocol for Joint Core Scheduling and Overload Control of μ s-Scale Tasks**
Bhaskar Pardeshi, Eric Stuhr, Ahmed Saeed *IEEE ICNP 2025*

Professional Experience

- Microsoft** Redmond, USA
Research Intern Jun 2026 – Aug 2026
 - Evaluated state-of-the-art load-balancing algorithms for multipath transport in SRv6-based AI training scale-out networks.
 - Designed a novel SRv6 network load-balancing algorithm that outperforms state-of-the-art approaches by achieving higher network utilization and lower queueing at switches, while remaining straightforward to implement on widely deployed NIC hardware.
- Nutanix, Inc.** San Jose, USA
Intern, Member of Technical Staff May 2025 – Aug 2025
 - Designed and implemented a delta-scan API for RocksDB, retrieving only changed key-value pairs after a base scan.
 - Improved scan efficiency, achieving up to 1000 $\times$ faster performance compared to the existing full-scan implementation.
- NVIDIA** Santa Clara, USA
GPU System Software Intern May 2024 – Jul 2024
 - Updated the user-space power, temperature, and acoustic controller for Nvidia GPU notebooks to accept inputs from a platform-specific service instead of ACPI.

- Ensured power draw remained within limits and reduced testing/debugging time by replacing SBIOS-based ACPI logic with user-space control.

- **VMware, Inc.** Bengaluru, India
Member of Technical Staff 2 Jan 2021 – Jul 2023
 - Created a simulated SmartNIC-based ESXi setup, reducing the need to procure expensive NVIDIA BlueField SmartNICs for testing.
 - Led a 3-person team to make ESXi's TCP/IP stack IPv6-compliant (RFC 8200/8201) and improved Cubic congestion control performance by 10–50% over NewReno, contributing to FreeBSD.
 - Enhanced virtual NIC to distribute incoming IPsec packets across multiple threads, scaling throughput proportionally.
 - Resolved 100+ TCP/IP and virtual network stack bugs and conducted knowledge transfer sessions for new engineers.

Projects

- **Userspace Multi-threading Library:** C library implementing the one-one, many-many, and hybrid userspace threading models.
- **Credit-based Scheduler for Userspace Threads:** Preemptive CPU scheduler for userspace threads, achieving proportional core allocation across threads.
- **Distributed Key-Value Store:** In-memory store with sharding and replication for high availability; maintained eventual consistency during host failures.
- **Userspace Asynchronous IPC Library:** Userspace asynchronous IPC library using shared memory and ring queues, with a high-level API for seamless communication with a local compression service.
- **User-space Write-Ahead-Log-based File System:** A persistent file system with crash recovery using an on-disk redo (write-ahead) log, including transaction APIs and a circular-queue log structure to ensure consistency.
- **Command-line Interpreter:** A UNIX command-line interpreter supporting process control, IPC, piping, I/O redirection, built-in commands, and job handling.

Technical Skills

- **Languages:** C, C++, Python, Go, Bash, JavaScript, SQL
- **Tools:** Git, Perforce, Make, GDB, Wireshark, gRPC, DPDK, RDMA, Redis, Memcached, RocksDB